

Bachelor Thesis Kick-Off

1. Agenda

- Introduction
- Research objective/questions
- Expected results and contributions
- Tools
- Timeline and Milestones

2. Introduction

Definieren von Zielen bzw. Meilensteine mit Kategorisierung von:

- mindestens notwendig
- sollte erreicht werden
- optional zu erreichen

Frage: BA in Englisch formulieren? Meetings auf Deutsch?

Context for Hensoldt: For Rust there is currently on-going development of tools and solutions. Ideally, we'd like to use some form of plug-in to allow special rules in the real-time data fusion.

TLDR: Developing a plugin system in Rust. As of right now, there is no ABI specification for Rust or similar

- Linux Kernelmodule über `insmod <file>`
- Java can load dynamic classes.

3. Research objective and questions

Goal of the paper is to analyze different approaches for developing a plugin system. The following points will be addressed:

- Illustrate the differences between compile time and dynamic time plugins with their respective advantages and disadvantages.
- How do the proposed options impact development time and complexity?
- Evaluate suitable tools and their inherent limitations, using specific examples such as WebAssembly (via interpreters), non-ABI-stable Rust formats, and others.

4. Expected results and contributions

- clear structured analysis of the current landscape for building plugin architectures in Rust
- there is no best solution, creating a "guide" instead for each approach
- provide prototypes

5. Some examples

Some things we can take a look at:

1. compile-time (using flags):
 - Cargo features
 - Conditional compilation
 - environment variables, external files, ...
2. runtime:
 - trait objects + boxed dyn types
 - enum + match
 - dynamic loading (.so, .dll) via libloading (C FFI)
 - ABI (non-)stable Rust formats (rlib, abi_stable crate)
 - embedded interpreters (WebAssembly, Lua)
 - hot-reloading from a config file (just changed parameters)
 - runtime flags by exposing environment variables

For each of these write about **safety, complexity, performance, limitations** and **interoperability**.

6. Tools

Work computer, virtual machine and work repository provided by the company.

NOTE: How can the Professor access the repository? Is a work Microsoft account necessary?

I'd like to use **Typt** (instead of LaTeX). What are the formal requirements for the document?

Google Scholar has many papers about Rust as a language.

7. Schedule

Month 0: Development of “base library” as a template for each approach. Write test suite for performance and correctness. Christmas/New-year will steal some time...

Month 1: Literature research, collection of various plugin architectures

Month 2: Analysis of the advantages and disadvantages, evaluation of the necessary libraries and tools and their impact on development time and complexity

Month 3: Implementation and evaluation of the architectures

Month 4: Interpretation of the results, derivation of recommendations for action, and writing of the bachelor thesis.